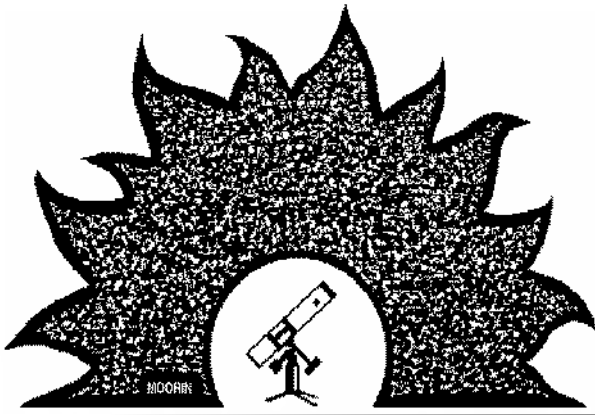




# PAStimes

Phoenix Astronomical Society

[www.pasaz.org](http://www.pasaz.org)

May 2018

Volume 69 Issue 9

PHOENIX ASTRONOMICAL SOCIETY — ESTABLISHED 1948

## Tom Polakis to Speak at May 10 Meeting

Provided by Tom Polakis

““CCD Photometry from Inside the Light Dome - Leisure Time Activity for a Retired Engineer”

In the past couple years, Tom Polakis has very actively pursued photometry with his CCD imaging equipment in Tempe. During that time, he has determined rotation periods of asteroids, measured brightness and colors of eclipsing binary and pulsating variable stars, and created light curves of exoplanets and an active quasar along the way. Much of this work has been or will be published in refereed journals. The first half of his presentation describes the fundamentals of photometry, his equipment and acquisition of images, and data reduction. Then the talk moves on to show a number of examples of his work that illustrate how measuring something as simple as the variation of light and color teaches us so much about stellar and Solar System astronomy.

Tom Polakis became an amateur as-

tronomer on the night of the 1977 Perseid meteor shower. He is a Contributing Editor for Astronomy Magazine, in which he has written many stories about the hobby and the profession. His interests are astronomical travel, visual observing, and all types of photography.”

**Terri**, Event Manager adds: The May PAS Meeting is being held this month on May 10th instead of the first Thursday of the month to accommodate the viewing possibilities at the Black Mountain Campus star party on May 3rd. We hope you can attend both events which are open to all.

Bring a friend, a snack to share and enjoy the evening with guest speaker Tom. Doors open 7pm, for socializing and the meeting begins at 7:30pm with a few brief announcements. We will have the PAS Officer Elections after which Tom will take the floor. Following Tom’s presentation will be the raffle, a few more announcements and adjournment of the

meeting.

Before the May 10th PAS meeting, PAS is having a Meet and Greet dinner for our speaker at Chili’s. If you want to attend, please RSVP by May 9th to [Events@pasaz.org](mailto:Events@pasaz.org) for the public and at this link: *See "May 10 Meeting" on Page 4*



*Photo of Tom Polakis provided by Tom.*

## PAS Meeting Review of Apr 5

By Terri, Event Manager

Prior to the April PAS meeting, a group of 13 PAS Members gathered at Chili’s for dinner. Attending were William, Terri, Sam, Rick, Don, Paul, John, Charles, Chris, Christine, Frank Gene (our guest speaker) and Debbie. Thank you all for joining us.

In the club President’s absence, the meeting was opened by Paul Facuna, Vice President, at 7:30. Our guest speaker was Gene Hanson, until recently director of the Milwaukee Astronomical Society. Gene did an entertaining presentation on the venerable history of MAS and shared a number of effective ways to help an astronomy club grow. At 8:50pm, member Sonny Schug

gave a talk on his astrophotography work and how he goes about it. Members also floated the idea that PAS should set up an Astrophotography SIG (Special Interest Group).

Past President Sam Insana had a few items to announce including PAS making a donation to the Grand Rapids Astronomical Association, the June 2nd PAS Field Trip to Northern Arizona and the upcoming ideas for the 70th PAS Anniversary Star Party. The meeting concluded about 9:40pm.

Many thanks to everyone who brought snacks to share. The case of water was

donated to PAS by Bruce Wurst. We hope to see you at the next PAS Meeting on May 10th.



*April PAS Meeting Guest Speaker Gene Hanson. Photo by Don.*

|                                   |                                                               |              |                                                                                          |
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| Phoenix Astronomical Society      | Mailing Address: 3039 W Peoria Ave #102-188, Phoenix AZ 85029 |              |                                                                                          |

## Upcoming PAS May Events

By Terri, Event Manager

Check out these and other fun events on the PAS Calendar: <http://www.pasaz.org/forums/calendar.php>. RSVP where requested to [Events@pasaz.org](mailto:Events@pasaz.org). On the PAS Calendar, "Private" means the event is for PAS members only. If you wish to become a PAS member and attend the private events, please download a membership form here: <http://www.pasaz.org/forums/downloads.php?do=file&id=129>

**May 1:** School Star Party (Private)

**May 3:** Huge Star Party at Black Mountain Campus 7pm to 9:45pm. RSVP requested. More details: <http://www.pasaz.org/forums/calendar.php?do=getinfo&e=2404&day=2018-5>

[3&c=1](#)

**May 4:** Private Star Party in Phoenix

**May 6:** Free Telescope Workshop at Bookmans 3:30 to 5:30pm. RSVP required. More details at this link: <http://www.pasaz.org/forums/calendar.php?do=getinfo&e=2415&day=2018-5>  
[6&c=1](#)

**May 10:** PAS Meeting at PVCC Main Campus in LS-203 at 7pm. More details: <http://www.pasaz.org/forums/calendar.php?do=getinfo&e=2392&day=2018-5>  
[10&c=1](#)

Dinner before the meeting for a meet and greet of the guest speaker at Chili's 4:30pm. RSVP required by day before.

More details: <http://www.pasaz.org/forums/calendar.php?do=getinfo&e=2568&day=2018-5>  
[10&c=1](#)

**May 11:** School Star Party (Private)

**May 11:** Arcosanti Star Party

**May 12:** Arcosanti Star Party

**May 13:** Arcosanti Star Party

**May 18:** School Star Party (Private)

At the time of completion of this issue, several more events were in the planning stages and could not be listed as "Scheduled" until they are definitely happening. Please see our online calendar for the events that didn't make it into this list.

## Summer 2018 PAS Events

By Terri Finch, Event Manager [Events@pasaz.org](mailto:Events@pasaz.org)

Check out these and other fun events on the PAS Calendar: <http://www.pasaz.org/forums/calendar.php>. RSVP where requested to [Events@pasaz.org](mailto:Events@pasaz.org). On the PAS Calendar, "Private" means the event is for PAS members only. If you wish to become a PAS member and attend the private events, please download a membership form here: <http://www.pasaz.org/forums/downloads.php?do=file&id=129>

During the summer months, the monsoon (Usually July through early September) season cancels many of the outdoor events PAS hosts. This is why RSVP is strongly requested for all events in July, August and September.

Events may be canceled with:

1) High winds of 10mph+

2) Any precipitation: If predicted above 20% chance of rain

3) Lack of telescope availability

4) Heavy cloud cover and/or 40%+ predicted cloud cover

**June 2:** PAS Field Trip (Private) for PAS Members to Flagstaff: <http://www.pasaz.org/forums/calendar.php?do=getinfo&e=2661&day=2018-6-2>

**June 3:** Free Telescope Workshop at Bookmans, 19th Ave and Northern in Phoenix - 3:30 pm to 5:30 pm. RSVP is required with Terri - [events@pasaz.org](mailto:events@pasaz.org). When you RSVP, email the make and model number of the telescope you need assistance with and that you are bringing to this event. We also assist with suggestions and advice on what telescope to purchase to match your needs. We have no telescopes

for sale at this event. More details: <http://www.pasaz.org/forums/calendar.php>

**June 23:** Bookmans Free Sidewalk Astronomy Event: <http://www.pasaz.org/forums/calendar.php?do=getinfo&e=2518&day=2018-6-23&c=1> Event is from 7:30 to 10pm.

**July 3:** Fireworks / Movie Night Party for PAS Members (Private): <http://www.pasaz.org/forums/calendar.php?do=getinfo&e=2532&day=2018-7-3&c=1>

**July 22:** Free Telescope Workshop at Bookmans, 19th Ave and Northern in Phoenix - 3:30 pm to 5:30 pm. RSVP is required with Terri - [events@pasaz.org](mailto:events@pasaz.org). When you RSVP, email the make and model number of the telescope you need assistance with and that you are bringing to this

# Summer 2018 PAS Events

event. We also assist with suggestions and advice on what telescope to purchase to match your needs. We have no telescopes for sale at this event. More details: <http://www.pasaz.org/forums/calendar.php?do=getinfo&e=2522&day=2018-7-22&c=1>

**July 28:** Bookmans Free Public Sidewalk Astronomy Event: <http://www.pasaz.org/forums/calendar.php?do=getinfo&e=2519&day=2018-7-28&c=1> Event is from 7:30 to 10pm.

**Aug 19:** Free Telescope Workshop at Bookmans, 19th Ave and Northern in Phoenix - 3:30 pm to 5:30 pm. RSVP is required with Terri - [events@pasaz.org](mailto:events@pasaz.org). When you RSVP, email the make and model number of the telescope you need assistance with and that you are bringing to this event. We also assist with suggestions and advice on what telescope to purchase to match your needs. We have no telescopes for sale at this event. More details: <http://www.pasaz.org/forums/calendar.php>

[?do=getinfo&e=2523&day=2018-8-19&c=1](http://www.pasaz.org/forums/calendar.php?do=getinfo&e=2523&day=2018-8-19&c=1)

At the time of completion of this issue of the newsletter, several more events were in the planning stages and could not be listed as "Scheduled" until they are definitely happening. Please see our online calendar <http://www.pasaz.org/forums/calendar.php> for more events that didn't make it into this list.

## Sandra Day O'Connor Star Party Mar 17

4 PAS members attended the STEAM event at Sandra Day O'Connor High School in Phoenix on Saturday night March 17th.

Rick Cunningham, Don Boyd, and Roger Anzini brought their telescopes to show the night sky to students, faculty and members of the public.

**Rick** states: "Estimated crowd tonight was 20 people. Very light attendance. Although the event started at 7:00 pm it wasn't until 7:50 pm when we had our first clients. They missed Venus and then Mercury setting over the mountains. I did show M42, Pollux double (Beta Geminorum), Mizar double (Zeta Uma) and one of the Cassiopeia doubles (Sigma Cassiopeiae). The seeing was good enough for excellent splitting of these three doubles. Since the crowd was so light I showed all three doubles to those interested."

Don viewed Venus, the Beehive cluster and the Pleiades. Roger had his 16 inch DOB trained on the spectacular Orion Neb-

ula all night. By Sam Insana

Sam Insana set up his meteorite collection and did a presentation for about 50 people who came by his table during the night. About half of the people were school volunteers so only about 25 were from the public. The school had several guest speakers in their auditorium and several displays in other buildings. These were all on the west end of campus. The dark area for the telescope viewing was on the east end about 300 yards away.

PAS appreciates being asked to be part of the school's fine program. The school had estimated that 400 to 500 people would attend. However, St Patrick Day parties put a big dent into that number according to some of the school organizers including our contact Dana Rasmussen. It was unfortunate that attendance was so low.

10 weather forecasts were mixed hours before the event. 4 showed a lot of clouds and possible rain, 2 reported some clouds, and 4 relatively clear. Once again the satellite photos, radar photos and barometer readings were most encouraging and they proved to be right.



Sam provided a super meteorite display at this school.  
Photo provided by Sam.

## May 15 2018

**Sunrise: 05:28**

**Sunset: 19:22**



**Q3: May 7**



**New: May 15**



**Q1: May 21**



**Full: May 29**

## What's Up For May?

By Don K. Boyd, PAStimes Editor

| Name           | Date            | Planets      |              |
|----------------|-----------------|--------------|--------------|
|                |                 | Rise         | Set          |
| Mercury        | 05-15-18        | 05:34        | 17:32        |
| <b>Venus</b>   | <b>05-15-18</b> | <b>07:21</b> | <b>21:47</b> |
| Mars           | 05-15-18        | 00:05        | 10:05        |
| <b>Jupiter</b> | <b>05-15-18</b> | <b>18:35</b> | <b>05:16</b> |
| Saturn         | 05-15-18        | 10:29        | 08:32        |
| Uranus         | 05-15-18        | 04:16        | 17:18        |
| Neptune        | 05-15-18        | 02:20        | 13:50        |
| Pluto          | 05-15-18        | 23:21        | 09:29        |

All Times Arizona Time

PAS is incorporated in the state of Arizona as a non-profit, scientific and educational 501(c)(3) organization. Our newsletter PAStimes is published monthly from September to May and distributed via the Internet. All Issues are available for download on our website [www.pasaz.org](http://www.pasaz.org). Ads for astronomy equipment are provided as a courtesy to sellers and buyers, and do not constitute any endorsement by PAS. All Photos by Don Boyd unless otherwise credited. All articles and photos are copyright their authors or PAStimes.

# May 10 Meeting

Continued from Page 1

<http://www.pasaz.org/forums/calendar.php?do=getinfo&e=2568&day=2018-5>

[10&c=1](#) for PAS members. Plan to arrive between 4:30 and 5pm. This is a BYO (Buy

Your Own) dinner which will end promptly at 6:30 to ensure a timely meeting start.

## Black Mountain Campus Star Party Mar 8

By Sam Insana & many attendees

**Sam** writes: I want to thank Mike Cawley, Rick Cunningham, Jenny Weitz and Mark Johnston for contributing to the extensive review below.

**Carol Murphy** writes: The kids loved the fair. Thanks for telling us about it.

**Tom McCarthy** writes: Thank you for having the event this past Thursday night. My students enjoyed looking through all different scopes and meeting your members. PVCC has a very nice up there and the skies were pretty dark.

Space is the place (Sun Ra)!

**Loretta Mondragon** writes: The event was well attended, from our clickers and those we know were there before the clickers. We are estimating 450-500 people attended!

**Stu Brackney** writes: As a "first timer" to the PAS night sky watch event last night...I must say you guys and gals do a very very nice event. I shall return and hope to make a meeting or two.

**Sam Insana** writes: All afternoon on March 8th the clouds looked threatening for the BMC star party. A few hours earlier I studied 10 weather forecasts predicting cloud cover averaging almost 60%. Normally I would have recommended that PAS cancel the telescopes. However, the satellite photos showed that most of the clouds had a good chance to go south of Cave Creek in the Valley. The satellites also showed likely gaps in the clouds during the night and that the clouds were very high altitude meaning they were thinner and it was possible to show bright stars, double stars and open

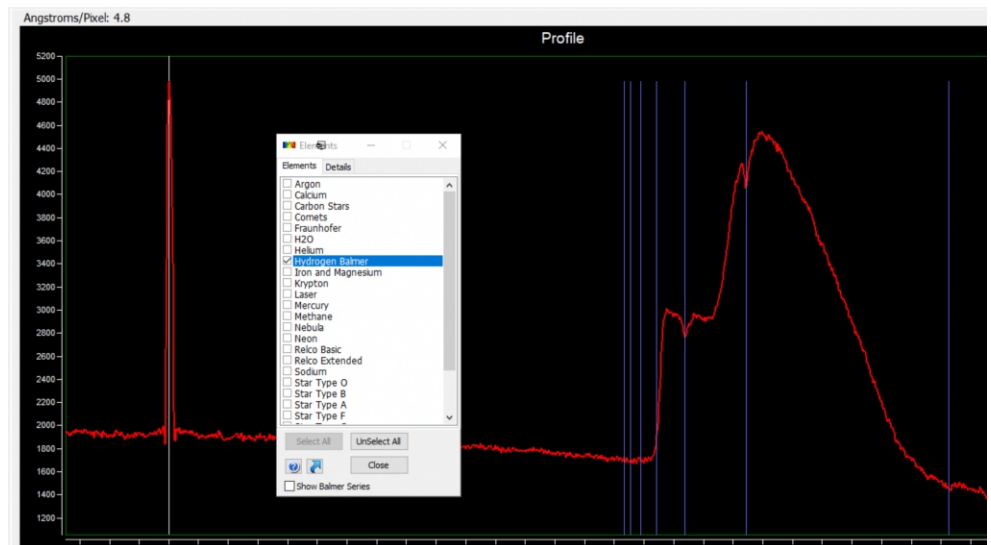
clusters. Since the barometric pressure had been rising all afternoon indicating that the weather might be getting a little better we decide to go ahead. We hadn't had a star party at BMC in several months and there was a large STEM event indoors to keep people busy if the clouds were bad. I suggested that a few PAS members may want to try the event, but if others wanted to stay home it would be understood. Four PAS members attended with their scopes: Rick Cunningham, Sam Insana, Mark Johnston and Mike Cawley.

**Mark** states "it was a cloud-dodging night, but ultimately very successful. While I had to switch objects frequently, there were no times when I wasn't able to show something in the sky to the guests at the event. I showed Venus, Mercury, Gamma

Andromeda, Miza/Alcor, Eta Cass, Hind's Crimson Star (carbon star), the Double Cluster, M46 (special request), and the Pleiades. There were lots of ooo's and ahh's. And as always the green laser was a hit as I pointed out constellations and asterisms. It was good we decided to show up because there were quite a few guests despite the iffy weather."

I am glad we decided to chance the event and it did turn out that looking at satellite photos is a big help on very questionable nights.

I want to thank Mike Cawley, Mike Marron, Rick Cunningham, Mark Johnston, and Ron Walker for attending what could have been a long dreary night but which turned out just fine.



Spectrographic analysis of a – star demonstrated at BMC star party. Photo and demo by Sam Insana.

## South Mountain Environmental Center March 24th

By Sam Insana

From 10 am to 2 pm, I provided my PST to show solar flares at the South Mountain Environmental Center. There were about 40 people who came up to look through my scope but only about six of them got to see a solar flare during the 15 minutes that clouds were not blocking the sun. The rest ended up enjoying my meteorites which I brought as a backup. I also handed out eclipse solar glasses to some of the children and showed their parents how to supervise them. The Center had other projects like constructing a sundial, doing solar cooking and several more interesting things.



Sam provided Solar Viewing through his PST until the sky was over run by clouds. Then he showed his Meteorite display to the public attending the South Mountain Solar Day on Mar 24. Photo provided by Sam.



# What's It Like Inside Mars?

By Jessica Stoller-Conrad

Mars is Earth's neighbor in the solar system. NASA's robotic explorers have visited our neighbor quite a few times. By orbiting, landing and roving on the Red Planet, we've learned so much about Martian canyons, volcanoes, rocks and soil. However, we still don't know exactly what Mars is like on the *inside*. This information could give scientists some really important clues about how Mars and the rest of our solar system formed.

This spring, NASA is launching a new mission to study the inside of Mars. It's called Mars InSight. InSight—short for Interior Exploration using Seismic Investigations, Geodesy and Heat Transport—is a lander. When InSight lands on Mars later this year, it won't drive around on the surface of Mars like a rover does. Instead, InSight will land, place instruments on the ground nearby and begin collecting information.

Just like a doctor uses instruments to understand what's going on inside your body, InSight will use three science instruments to figure out what's going on inside Mars.

One of these instruments is called a

seismometer. On Earth, scientists use seismometers to study the vibrations that happen during earthquakes. InSight's seismometer will measure the vibrations of earthquakes on Mars—known as marsquakes. We know that on Earth, different materials vibrate in different ways. By studying the vibrations from marsquakes, scientists hope to figure out what materials are found inside Mars.

InSight will also carry a heat probe that will take the temperature on Mars. The heat probe will dig almost 16 feet below Mars' surface. After it burrows into the ground, the heat probe will measure the heat coming from the interior of Mars. These measurements can also help us understand where Mars' heat comes from in the first place. This information will help scientists figure out how Mars formed and if it's made from the same stuff as Earth and the Moon.

Scientists know that the very center of Mars, called the core, is made of iron. But what else is in there? InSight has an instrument called the Rotation and Interior Structure Experiment, or RISE, that will hopefully help us to find out.

Although the InSight lander stays in one spot on Mars, Mars wobbles around as it

orbits the Sun. RISE will keep track of InSight's location so that scientists will have a way to measure these wobbles. This information will help determine what materials are in Mars' core and whether the core is liquid or solid.

InSight will collect tons of information about what Mars is like under the surface. One day, these new details from InSight will help us understand more about how planets like Mars—and our home, Earth—came to be.

For more information about earthquakes and marsquakes, visit: <https://spaceplace.nasa.gov/earthquakes>

## Bookmans Telescope Workshop Apr 8

By Terri, Event Manager

PAS received 3 RSVP's for this event, two of whom were able to attend, along with one walk-in. Robert came with his Unitron 114 60x900 refractor. He arrived at the beginning of the event and stayed the full 2 hours initially being helped by Sam Insana and Bruce Wurst. Many thanks to Rodney Fong for helping to transport the scope into Bookmans and then back to Robert's vehicle. After receiving help from Sam and Bruce Wurst, he sat at the PAS Table and asked for advice on what telescope to purchase from William and Terri Finch.

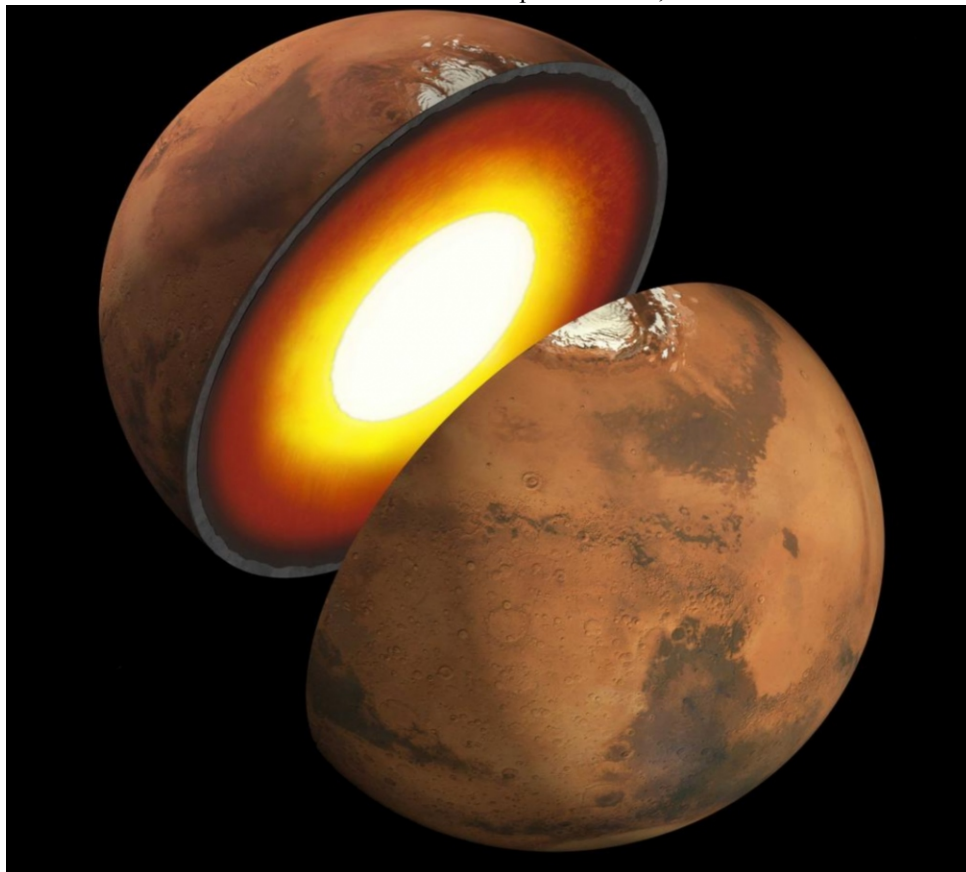
We also had the pleasure of meeting Brett and Thomas who came to find out what telescope to purchase. William and I assisted him along with Sam and Rick Cunningham. Brett and Thomas left about 4:45pm with tons of information and notes. We hope to see them again.

We also had the company of PAS Members Ed and Bette Wurst at this event.

The event concluded at 5:30 after which, several PAS members plus Donna and Helen Zander had dinner at Carlos O'Briens.

Many thanks to the telescope team for their time and energy, and to Bookmans Entertainment Exchange for lending us the space at the front of the store to hold this event.

Looking forward to the next Free Telescope Workshop at PVCC and Bookmans: Apr 19, May 6, June 3, July 22, and July 19. Please always check the PAS online Calendar to be sure the event is happening <http://www.pasaz.org/forums/calendar.php>.



An artist's illustration showing a possible inner structure of Mars. Image credit: NASA/JPL-Caltech

# Hidden Hills Star Party Mar 23

By Terri, Roger, and Rick

**Roger** writes: Several months before our Hidden Hills event took place I suggested to Mrs. Rea, my daughter's 5th grade science teacher that she might want to consider a stargazing event with PAS. It took no more than 10 seconds for her to become excited about the prospect of a star show for her students. Our event day dawned to a valley-wide layer of overcast that eventually gave way to the surprise of clearing skies.

The event itself began in the afternoon with an imaginary trip of 1,350 light years to the Orion Nebula, getting everyone there and back before the final bell rang.

Later our PAS team consisting of Terri, Storm, Bruce, Sam, Rick, William and me set up for our evening program. Sam set up nearby with his meteorites as our wonderful liaison and hostess, Joy greeted our team about 5:45 PM with food, drinks and general help.

We had a large, enthusiastic group of over 70 parents and students viewing the Moon, Venus, Orion Nebula and a number of additional objects. Our PAS SkyTours team loves sharing our scopes and knowledge with anyone interested in our fantastic cosmos. We are already looking forward to next seasons event, until then, good health and clear skies to all.

Our turnout was one of the best events ever for their school - among the observers were both fifth grade teachers, Mrs. Rea and her family and Mr. Botello with his entire family. I want to say a special thank you to both of these wonderful teachers on behalf of my daughter Sonya and myself: they have made a real difference in Sonya's life.

**Terri** writes: Thanks to the telescope team for helping to make this a super event. Joy was thrilled and PAS plans to do another event at this school in for Fall 2018. Thanks also to Joy for having PAS at her school. Photographs taken for the school website are also included at this link for your enjoyment:  
<https://www.dropbox.com/sh/7a1gpzxdyvrnx06/AADmB4GF1Igr8acgQ1N4NqBha?dl=0>

**Rick** writes: I showed Venus, M-42, M-43, Mizar double and Pollux double. The ISS also was visible at 8:07 for several

minutes. After the ISS Passover the event quickly ended. Estimating 50-70 people during the evening, a good mix of parents and students.

**Sam** writes: March 23rd, I attended the PAS star party at Hidden Hills elementary school. While other PAS members handled their telescopes, I set up and ran the meteorite demonstrations. I had 20 meteorites, 2 tektites, charts, posters, and other items to show the students, parents and faculty. About 70 people came by my table to talk and check out the items.



*It was a clear night for doing a star party. We had a great turnout and we started before dark. Photo by Terri.*



*Bruce was set up near Roger and was seen sitting on his step ladder while watching the presentation Roger provided. Photo by Terri.*



*Rick was seen looking to the sky for the first signs of Venus. Storm was grabbing an image of the Moon while William assisted her. Photo by Terri.*



*Headless Storm is looking for the Moon through her telescope with William behind her scope in this photo. Terri's scope is to the right. Photo by Terri.*

# More Summer 2018 Public Events

By Terri, Event Manager

PAS is putting together several public star parties and lectures in Carefree this summer. These events are free and open to the public of all ages. We request that you RSVP at [Events@pasaz](mailto:Events@pasaz) with the number in your party so we can plan for attendance (seating is limited) and so we can let you know if an event is canceled. All of the events, whether they are a lecture or a star party are weather permitting and all will have potlucks around dinner time and then the lecture and/or star party begins. Please monitor the [PASaz.org](http://PASaz.org) website calendar for

updates. Here are the tentative dates for these events:

**Public Star Party May 19:**  
<http://www.pasaz.org/forums/calendar.php?do=getinfo&e=2679&day=2018-5-19&c=1>,

**Public Star Party June 16:**  
<http://www.pasaz.org/forums/calendar.php?do=getinfo&e=2680&day=2018-6-16&c=1>,

**Public Lecture June 30:**  
<http://www.pasaz.org/forums/calendar.php?do=getinfo&e=2677&day=2018-6-30>

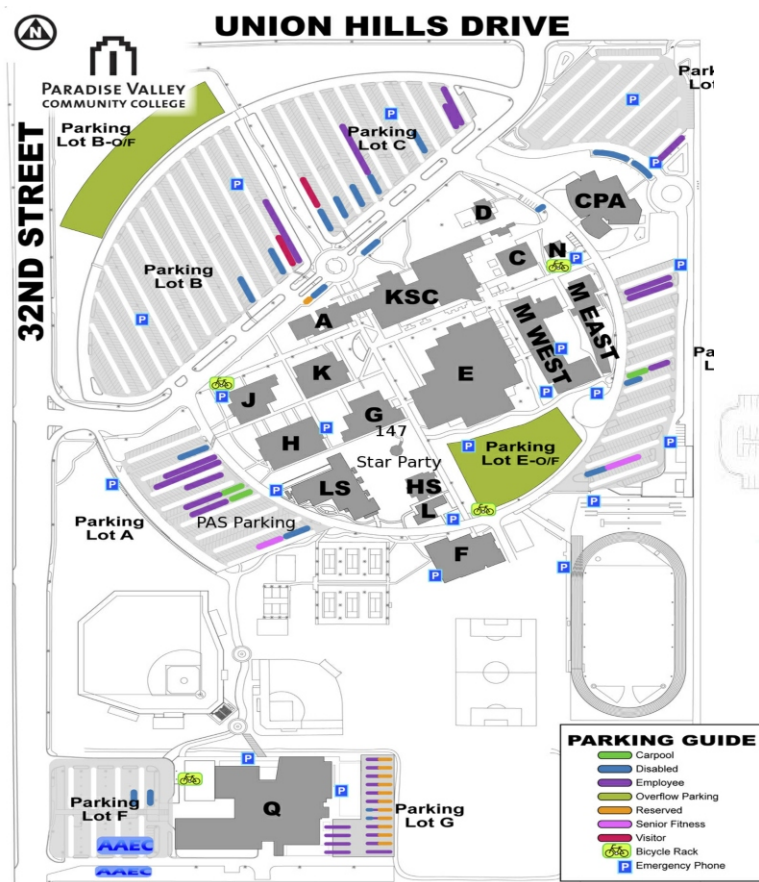
[30&c=1](http://www.pasaz.org/forums/calendar.php?do=getinfo&e=2681&day=2018-7-14&c=1),

**Public Star Party July 14:**  
<http://www.pasaz.org/forums/calendar.php?do=getinfo&e=2681&day=2018-7-14&c=1>,

**Public Lecture June 21:**  
<http://www.pasaz.org/forums/calendar.php?do=getinfo&e=2678&day=2018-6-21&c=1>. Please note: the topics for the lectures are in the process of being chosen and will be announced on the calendar page for that lecture date. We would love for you to attend.

## Map of PVCC Main Location

18401 N. 32nd Street | Phoenix, AZ 85032



## Map of PVCC Black Mountain

34250 N. 60th Street | Scottsdale, AZ 85266



## 2018 PAS GUEST SPEAKER LINE-UP

By Terri Finch, Event Manager [Events@pasaz.org](mailto:Events@pasaz.org)

PAS Meetings are held at PVCC Main Campus on the first Thursday of the month in Room: LS-203 unless otherwise noted. Do you have an idea for a guest speaker? Email Terri the details. Here are the next scheduled guest speakers at PAS:

\*\*\* Summer Break - No General Meetings \*\*\*

**Sep 6:** Laurence Garvie, Topic: Meteorites

**Oct 4:** Tom Mozdzen, Topic: EDGES project (Experiment to Detect the Global Epoch of reionization Signature)

**Nov 1:** Vincenzo Cataldo, Topic: Planetary Volcanism

More information at:

<http://www.pasaz.org/index.php?pageid=meetings>

## Scope for sale:

Hand Built by Chet Schuler, PAS member. It's a well made 4" scope. Asking \$50 OBO. The sale of the scope will be a donation to PAS. Pick up and viewing of telescope is in the ASU West area. Call Kevin at 602 795 7440.

## Thanks From President Mike

I want to thank all the members of PAS for their kind thoughts and condolences on the passing of my father Jim Marron. He

was an inspiration to generations of people in both amateur astronomy and acting. An encomium and obituary can be found with a

"Jim Marron" search or you may go to <https://www.youtube.com/watch?v=e4UfNC1DKe4> for video. President Mike Marron.